Master Thesis

Configuration-based mobile applications

zur Erlangung des akademischen Grades

Master of Science

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von

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Erklärung der Selbstständigkeit

Hiermit versichere ich, die vorliegende Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie die Zitate deutlich kenntlich gemacht zu haben.

Gießen, den February 7, 2019 Samuel Schepp

This thesis describes configuration-based mobile applications and their infrastructure. In addition, a possible way of implementing a configuration system is discussed. Finally, there is an evaluation of the use case of interpreting and generated apps.

Furthermore, the implementation is compared to the approach of established literature by classifying the model and interpreter.

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1 Introduction

Mobile application and digital media usage is still on the rising[Per]. Apps are one of many possible ways to interact with customers and consumer. A web appearance on social media or a dedicated web page can also fulfill these needs. The way, the user interacts with a web page is more shallow, because of lower expectations. A user needs to log in most of the time.

A mobile application can enable user specific features without logging in, because the user can be identified with an identification token, that is generated, while the user installs the app. It is possible to activate special features like geofencing, push notifications or local data storage, that is synchronized with the platform cloud services (e.g. Apple iCloud or Google Drive).

An mobile app integrated more deeply into the device, than a web app. However, web browser already support a lot of the features by now, that originally are only available to native apps. The current development indicates, that the differences will be further eliminated by the browser vendors.

1.1 Problem definition

Mobile applications have significant disadvantages, that push the development of web apps. Changes have to be approved by the phone vendors, in order to be able to release update to the app. This causes development delays and reduces the ability to react to events or customer requirements.

In addition, managing a lot of apps (> 20) requires a lot of work, since all of them have to be maintained. New operating system features have to be implemented in all of the apps, which can cause inconsistencies. All apps have to be tested in detail, to make sure that all apps implement the required change correctly.

Developers need deep system knowledge to develop the apps. Every change has to go through one of each platform experts, so development is expensive.

1.2 Motivation

use of native mobile features, camera, background gps, beacons, push notification (with payload). caching, fast ui, good battery life.

no need to depend on third party frameworks, direct use of phone vendor's apis.

1.3 Distinction

This thesis will not describe the development experience nor the initial expense of the project. Neither will this thesis evaluate financial aspects and at which point, the project was or is worth the development costs.

1.4 Structure

1.4.1 Research Question

The thesis aims to answer the question "Can a code interpreter be integrated in any domain?". In addition, the development approach is compared to procedu- res, that are described in popular literature.

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2 Background

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2.1 Fabrik19

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2.2 Mobilitysuite

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3 Concept

3.1 Model driven development

3.1.1 Code generators

3.1.2 Code interpreters

3.2 Implementation goals

Same day delivery

easy configuration

simple datasources

common app tasks simplified

content driven (wordpress, social media)

reactive form elements

3.3 Trade offs

boundary conditions

limits to development of app contents (app is out, but the customer needs more functionality, which is not possible: incomprehensible reasons for limits.)

4 Implementation

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4.1 Components

4.1.1 Nodes

4.1.2 Layouts

4.1.3 Database

4.1.4 Actions

4.1.5 Data proxy

4.2 Infrastructure

4.2.1 Control center

4.2.2 Resource server

4.2.3 Image cruncher

5 Evaluation

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5.1 Analysis

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5.1.1 Benefits

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5.1.2 Disadvantages

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5.2 Classification

5.2.1 Approach

5.2.2 Model

5.3 Conclusion

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Bibliography

[Per] Perkins, Kleiner: Internet Trends Report 2018, URL https://www.kleinerperkins.com/perspectives/internet-trends-report-2018/

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